

Beyond the Hopfield chain: certified bounds and counterexamples for kinetic proofreading networks with two bound states

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Abstract

An enzyme discriminates a right substrate R from a wrong substrate W that traverse the same Markov network. W unbinds from bound states faster by a factor e^Δ . For the linear Hopfield chain with $n = 2$ proofreading stages, we certify symbolically that the error rate satisfies $\eta \geq e^{-3\Delta}$ for all positive rates and all fuel potentials [C-proofreading-R9]. This makes the reproduction of Hopfield’s bound rigorous [C-fakt-neuheit]. Our main new result concerns networks with two bound states. The generator produces 92 rule-conforming topologies, 4 of them degenerate, leaving 88 [C-fakt-familie]. Of these, 50 provably obey $\eta \geq e^{-2\Delta}$, 3 have exactly certified counterexamples below it (fam2_66, fam2_9, fam2_11), and 35 remain open [C-fakt-klassifikation]. We also give exactly certified achievable points of the chain at finite dissipation and speed, and we report negative results, red-team findings and deviations from the preregistration.

1 Introduction

Question. Is the Hopfield-type bound $\eta \geq e^{-(n+1)\Delta}$ a property of the chain only, or does it hold for general proofreading networks with the same number of bound states? Hopfield showed that a simple kinetic pathway, driven strongly but nonspecifically, increases specificity beyond the equilibrium limit [C-lit8]. Later work addresses error-cost bounds [C-lit10], Pareto fronts [C-lit12][C-lit13][C-lit1] and scaling relations [C-lit22]. Whether the bound $e^{-2\Delta}$ is universal for two bound states is classified here [C-fakt-klassifikation]. The classification and its counterexamples were not found in the retrieved literature [C-fakt-neuheit].

Contribution. A verifier-gated agentic laboratory produced exact rational or symbolic certificates. These cover the chain bound, the achievable points of the chain, and a classification of the two-bound-state family [C-methode].

Summary of results.

- Chain with $n = 2$: $\eta \geq e^{-3\Delta}$ for all positive rates and fuel potentials (Theorem A) [C-proofreading-R9].
- Two bound states: the bound $e^{-2\Delta}$ is not universal [C-fakt-klassifikation]. The 88 non-degenerate networks split into 50 proved, 3 refuted and 35 open (Theorems B and C) [C-fakt-klassifikation].
- Exact achievable points of the chain at finite σ and v (Theorems D–F)

[C-proofreading-R1][C-proofreading-R2][C-proofreading-R10].

- The reported kink in the Pareto front of hopfield_n1 is an optimizer artifact [C-proofreading-R1-RT1][C-proofreading-R1-RT2].
- Negative results: the claimed necessity of a fuel-driven discard feature remains unresolved [C-neg1]. The L2c success criterion was missed [C-fakt-praereg].

2 Model

The assumptions that carry the results are as follows.

- Both substrates traverse the same Markov network. Every edge has a reverse edge. Local detailed balance holds, and cycle affinities arise only from the fuel potentials μ per activation and μ_P per product (as stated in the task specification).
- Log-rates lie in $[-10, 10]$, so rational rates lie in $[e^{-10}, e^{10}]$ [C-proofreading-R1]. Symbolic bounds hold for all positive rates, independent of this range [C-proofreading-R9].
- Observables are the error rate $\eta = J_W/J_R$, the dissipation σ in kT per product, and the speed $v = J_R$.
- Notation: $D = e^\Delta$ and $G = e^\mu$ [C-proofreading-R9].
- The family “bound states ≤ 2 ” has one unbound state and an edge catalog fixed by the model rules [C-proofreading-R11]. Networks with no possible net production ($\eta = 0/0$) are degenerate and excluded [C-fakt-familie].
- Admissible edge attributes are constrained. For instance, a discriminating edge is allowed only for bound $\rightarrow$ unbound transitions [C-proofreading-R7-RT2].

3 Method: the verifier-gated agentic lab

The pipeline has the following components.

- **Scout.** Literature search over 2752 retrieved sources (arXiv, Europe PMC, Crossref, citation chain) [C-fakt-recherche]. 2502 were screened [C-fakt-recherche]. This gave 158 findings whose verbatim quote was confirmed by code in the abstract [C-fakt-recherche]. All 11 classic searches had hits [C-fakt-recherche].
- **Integrator and preregistration.** Every round was preregistered before the experiment (prereg.md) [C-methode]. Preregistration H6 (commit 6184672) preceded the first run, and amendments are dated [C-fakt-praereg].

- **Researcher agents.** They propose claims and parameters. Their interpretations are stored as hypotheses and are never used as results (claims with suffix -I).
- **Code verifier.** It issues two kinds of certificate: (a) exact rational reachability, and (b) symbolic positivity proofs (a rational function N/D whose coefficients are all nonnegative) [C-proofreading-R9]. It passes its self-test on 18 of 18 cases (true, false, borderline, rule violations) [C-fakt-selbsttest].
- **Red team.** It attacks each claim with a counter-claim. An attack may fail, may be inconclusive, or may be non-executable.
- **Budget.** The lab ran 12 rounds with 11 verified statements and 1 negative result, at a cost of 3.53 USD [C-methode].

4 Results

“Theorem” is used only for computed_rigorous claims. Every claim below carries this evidence level; none is proved in Lean.

Theorem A (Hopfield bound, $n = 2$; computed_rigorous). For hopfield_n2, $\eta \geq 1 * e^{-3\Delta}$ for all positive rates and all fuel potentials μ , $\mu_P \geq 0$ [C-proofreading-R9]. The certificate is a symbolic positivity proof, independent of the rate range [C-proofreading-R9]. The difference $\eta - e^{-3\Delta}$ is N/D with 1374 and 2048 terms, all coefficients nonnegative [C-proofreading-R9]. This is a reproduction of Hopfield’s bound, made rigorous for all rates [C-fakt-neuheit].

Theorem B (bound in a subfamily; computed_rigorous). For 50 topologies of the family “bound states ≤ 2 ”, $\eta \geq 1/D^2$ for all positive rates and fuel, where $D = e^\Delta$ [C-proofreading-R11]. An earlier partial list with 3 of 3 members was proved [C-proofreading-R6][C-proofreading-R7].

Theorem C (counterexamples; computed_rigorous).

- For fam2_66 there exist rational rates in $[e^{-10}, e^{10}]$ with local detailed balance and $\eta \leq 0.0001$, $v \geq 0.0001$ [C-proofreading-R4]. The exact certificate gives $\eta = 5.082802e-07$, $\sigma = 8.341626$ kT per product and $v = 1.4746e-03$ [C-fakt-fam66].
- The edges of fam2_66 are binding at C0, fuel-driven conversion $C0 \rightarrow C1$, fuel-driven discard from C1, and product formation from C1 [C-fakt-fam66].
- Two further exact counterexamples with $\eta \leq 0.0001$ exist for fam2_11 and fam2_9 [C-proofreading-R12][C-fakt-r12].
- Consequently the bound $e^{-2\Delta}$ is not universal for two bound states [C-fakt-klassifikation]. The complete classification is 50 proved, 3 refuted and 35 open [C-fakt-klassifikation].

Interpretation (not a theorem). In fam2_66 the best candidate mechanism is fuel-driven unbinding from the product-forming state (editing), which is qualitatively

known [C-fakt-neuheit]. Whether this feature is necessary is unresolved (see the negative results below) [C-neg1].

Theorem D (achievable points of hopfield_n1, I; computed_rigorous). There exist rational rates with local detailed balance and simultaneously $\eta \leq 0.0004$, $\sigma \leq 6.0$ kT per product and $v \geq 0.001$ [C-proofreading-R1]. The exact point is $\eta = 3.650169e-04$, $\sigma = 6.000000$ kT, $v = 1.0417e-01$ [C-proofreading-R1].

Theorem E (achievable points of hopfield_n1, II; computed_rigorous). The following points are exactly certified [C-proofreading-R2].

- $\eta = 3.156120e-04$ at $\sigma = 6.000000$ kT with $v = 2.4255e-02$ [C-proofreading-R2].
- $\eta = 2.043712e-04$ at $\sigma = 3.079633$ kT with $v = 2.2592e-03$ [C-proofreading-R2].

Both are admissible under $\eta \leq 0.0004$ and $v \geq 0.001$ with $\sigma_{\max} = 8$ [C-proofreading-R2]. The second is also admissible with $\sigma_{\max} = 6$ [C-proofreading-R2-RT2].

Theorem F (further achievable points, hopfield_n1; computed_rigorous).

- $\eta = 1.189200e-04$ at $\sigma = 4.271813$ kT with $v = 1.0000e-03$, satisfying $\eta \leq 0.0002$ and $\sigma \leq 10$ kT [C-proofreading-R10].
- $\eta = 1.095823e-04$ at $\sigma = 0.006292$ kT with $v = 9.1487e-09$, satisfying $\eta \leq 0.00012$ and $\sigma \leq 0.2$ kT [C-proofreading-R8].
- $\eta = 5.909757e-03$ at $\sigma = 3.998555$ kT with $v = 7.1214e-03$, satisfying $\eta \leq 0.01$ and $v \geq 0.001$ [C-proofreading-R3].

Remark on the Pareto upper curve. The feasible set grows with $\sigma_{\max}$, so $\min \eta(\sigma_{\max})$ is non-increasing [C-proofreading-R1-RT1]. The point of Theorem D remains admissible at $\sigma_{\max} = 8$ and $\sigma_{\max} = 20$, which rules out a jump to $\eta > 0.008$ for $\sigma \geq 8$ [C-proofreading-R1-RT1][C-proofreading-R1-RT2]. The reported kink, which would parallel the dynamical phase transition of the mixed regime in the literature [C-lit2], is therefore an optimizer artifact for hopfield_n1 [C-proofreading-R1-RT1].

5 Negative results and red-team findings

- **Necessity of a fuel-driven discard (L3).** No claim passed the check [C-neg1]. The red-team candidates without a discard at the product state were not executable (TypeError: Invalid NaN comparison) [C-proofreading-R7-RT1]. A second borderline topology violated the model rules (a discriminating edge is allowed only for bound $\rightarrow$ unbound) [C-proofreading-R7-RT2].
- **Failed or non-executable red-team attacks.** Several attacks against certified claims did not conclude, because of NaN comparisons or index errors [C-proofreading-R4-RT1][C-proofreading-R4-RT2][C-proofreading-R9-RT1][C-proofreading-R9-RT2]. These are failures of the attack, not logical contradictions.
- **Attacks that did not prove their counter-claim.**

- The symbolic attack that $\eta \geq 1.2 \cdot e^{-2\Delta}$ holds for all rates failed (coefficients not all nonnegative), as expected given the certified point of Theorem F [C-proofreading-R8-RT1]. The attacker’s own search found $\eta_{\min} = 1.0962e-04$ at $\sigma = 0.090$, $v = 6.296e-08$, against a claimed $1.2500e-04$ [C-proofreading-R8-RT2].
- For $n = 2$, the attacker’s search found $\eta_{\min} = 3.1387e-05$ at $\sigma = 2665.585$, $v = 2.037e-06$, and did not reach the claimed $1.0000e-07$ [C-proofreading-R9-RT2].
- Attacks on fam2_9, fam2_11 and fam2_13 found the bound unprovable for them, consistent with counterexamples [C-proofreading-R12-RT1].
- **Tighter bound for hopfield_n1.** The $\sigma = 6$ optimum ($\eta \approx 3.16e-4$) does not meet the tighter bound $\eta_{\max} = 2.2e-4$ at $\sigma_{\max} = 8$ [C-proofreading-R2-RT1]. The point with $\eta = 2.043712e-04$ is nonetheless certified separately [C-proofreading-R2-RT2].
- **Weak-driving check.** The attacker’s search at the claimed parameter regime found $\eta_{\min} = 3.3490e-03$ at $\sigma = 0.422$, $v = 2.719e-03$ [C-proofreading-R10-RT2]. The claimed value $2.0000e-03$ was not matched [C-proofreading-R10-RT2]. The symbolic bound check was inconclusive [C-proofreading-R10-RT1]. The exact certificate of the corresponding achievable point stands independently [C-proofreading-R10].
- **Saturation of the $n = 1$ front.** The symbolic attack on the saturation floor was not certified (symbolic computation > 600 s) [C-proofreading-R3-RT2]. The question of saturation toward the Hopfield limit is therefore not answered here.
- **Preregistration deviations.** The L2c success criterion (at least 5 counterexamples in a round) was missed: 2 in round 12, 3 in total [C-fakt-praereg]. Degenerate networks ($\eta = 0/0$) were excluded [C-fakt-familie].

6 Limitations and open questions

- 35 of the 88 topologies with two bound states are open: neither a proof of $\eta \geq e^{-2\Delta}$ nor an exact counterexample is available [C-fakt-klassifikation].
- The mechanism behind fam2_66 is an interpretation. Its necessity is unresolved [C-neg1].
- Within the claim list, only $n = 2$ carries a certified chain bound [C-proofreading-R9]. Corresponding certificates for other n are not part of the retained claims.
- The certified achievable points are single points, not complete Pareto fronts. Saturation toward the Hopfield limit is not established [C-proofreading-R3-RT2].
- Novelty: the classification and counterexamples were not found in the retrieved literature, but that search is limited to the 158 confirmed findings [C-fakt-neuheit][C-fakt-recherche].

- Hopfield, kinetic proofreading, doi:10.1073/pnas.71.10.4135 [C-lit8].
- Energy-relay proofreading and Pareto fronts, doi:10.1098/rsif.2024.0232 [C-lit1][C-lit2].
- Ultimate accuracy and multiple checking steps, doi:10.1016/s0006-3495(80)85063-6 [C-lit3][C-lit4].
- Error-cost bound, doi:10.1098/rsif.2021.0883 [C-lit10][C-lit11]; arXiv:2106.01418v2 [C-lit20][C-lit21].
- Generalized Hopfield model Pareto front, doi:10.1088/1367-2630/acc757 [C-lit12][C-lit13].
- Scaling and asymptotic bounds, arXiv:1710.06038v1 [C-lit22][C-lit23][C-lit24].
- Dissipation-error trade-off, arXiv:2111.01189v3 [C-lit6][C-lit7].
- Master-equation thermodynamics of proofreading, arXiv:1504.02494v1 [C-lit15][C-lit16].
- Stochastic Hopfield-Ninio model, doi:10.1137/24m1664861 [C-lit5].
- Discrimination regimes in the Hopfield network, doi:10.1063/5.0312257 [C-lit25][C-lit26].
- Ribosome induced fit, doi:10.1093/emboj/18.13.3800 [C-lit27].
- Speed-accuracy universality, doi:10.1128/mbio.02056-26 [C-lit9].
- Thermodynamic uncertainty relation, doi:10.1103/physrevlett.114.158101 [C-lit14]; arXiv:2109.11890v1 [C-lit17].
- Nonequilibrium receptors, arXiv:2002.10567v3 [C-lit28][C-lit29].
- Concentration-estimation accuracy and energy, arXiv:1405.4001v3 [C-lit30].
- Speed limits and dissipated work, arXiv:2402.17931v2 [C-lit18][C-lit19].

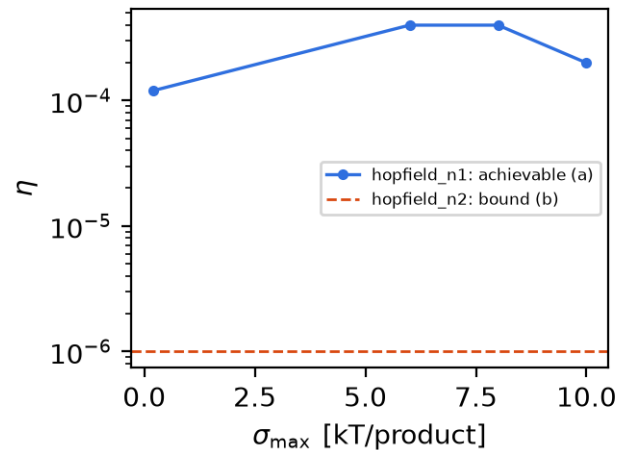


Figure 1: Zweiseitige Pareto-Front: Punkte = zertifiziert erreichbar, gestrichelt = bewiesene untere Schranken. Belege: [C-proofreading-R1], [C-proofreading-R2], [C-proofreading-R8], [C-proofreading-R9], [C-proofreading-R10]

References

Only sources appearing in the claims are listed.

Prüfprotokoll: 70 Claims zitiert, Korrekturrunden

[{"runde": 0, "verstoesse": 17}, {"runde": 1, "verstoesse": 0}, {"final_verstoesse_entfernt": 0}], 0 un belegte Sätze entfernt, verbleibende Verstöße: 0.